Climate Policy and Structural Change: Evidence from Energy Transitions in Latin America.

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IFORS 2026

Agenda

1. Introduction
2. System Modeling
3. Data & Causal-Inference Design
4. Results
5. Conclusions
6. Annex

Outline

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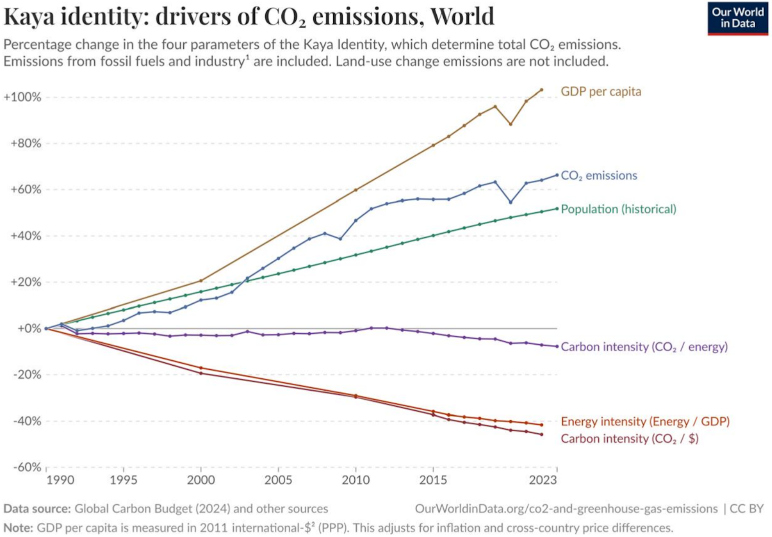
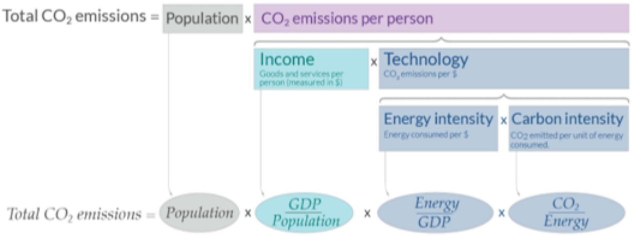
Mechanisms for Decoupling: Reducing CO2 emissions while growing



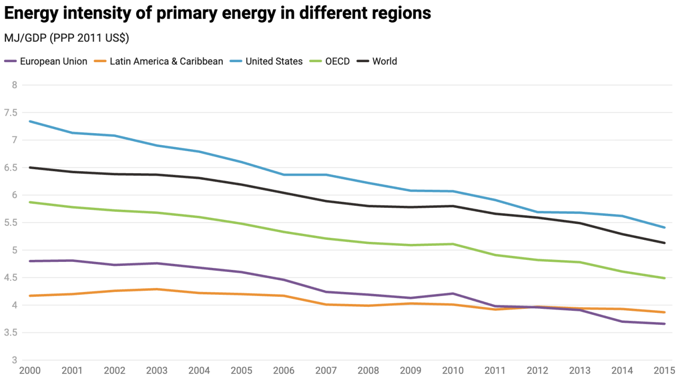
What are the drivers that determines the CO2 emissions?



Decarbonization *-reductions in Energy Intensity-* can be framed as the optimization of two subsystems:

- **Technique effect:** the internal efficiency of each productive sector.
  - adopting energy-saving technologies,
- **Composition effect:** the reallocation of resources across sectors.
  - transitioning to less energy-intensive industries.

Latin America faces unique challenges in decarbonization:



- High reliance on fossil fuel exports.
- Significant emissions from land use change and agriculture (LULUCF).
- High vulnerability to climate change impacts.
- Socio-economic factors: Informality, inequality, fiscal dependence on resources.
- **Carbon pricing, used globally as a market-based mechanism to steer both channels.**

Research Questions and Objectives

Research Questions:

- What are the main drivers of the long-term decline in energy intensity in Latin America?
- What is the effect of a carbon-tax intervention on the two optimization channels—sectoral efficiency and structural reallocation—in Latin America?

Research Questions and Objectives

Research Questions:

- What are the main drivers of the long-term decline in energy intensity in Latin America?
- What is the effect of a carbon-tax intervention on the two optimization channels—sectoral efficiency and structural reallocation—in Latin America?

Objectives:

1. Develop a theoretical model of how energy prices affect energy intensity via the technique and composition channels.
2. Apply an additive Fisher index decomposition to document the historical drivers of energy intensity across 21 Latin American countries, 1990–2020.
3. Identify causal effects using a Synthetic Control Method (quasi-experimental design).

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Model Formulation: A Constrained Multi-Sector Optimization Problem

Basic set-up:

- A multi-sector system: a representative consumer and several sectors producing with labor, capital, and energy.
- Capital and energy are *perfect complements* (Leontief); this composite combines with labor via Cobb-Douglas.
- The energy price is an exogenous parameter, shifted upward by a policy measure.

Model Formulation: A Constrained Multi-Sector Optimization Problem

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- The energy price is an exogenous parameter, shifted upward by a policy measure.

- Sector-i firms produce output $Y_{i,t}$ minimizing costs under a strict physical constraint: capital (K) and energy (e) are perfect complements (Leontief technology):

$$Y_{i,t} = A_{i,t} \left(\min[K_{i,t}, \gamma_{i,t} e_{i,t}] \right)^{\alpha} L_{i,t}^{1-\alpha}$$

  where $\gamma_{i,t}$ is the endogenous utilization efficiency of capital.
- Induced Technological Innovation (Assumption 2): $\gamma_{i,t} = g_i(p_{e,t})$ is endogenous to exogenous price shocks ($g_i'(\cdot) > 0$).

see more

Endogenous Efficiency: Induced Innovation in the Model

General Equilibrium State Variable (Aggregate Energy Intensity):

$$\frac{E_t}{Y_t} = \alpha \sum_{i=1}^{N} \frac{\sigma_{i,t}}{g_i(p_{e,t})r_t + p_{e,t}} \quad (1)$$

Energy consumption per unit of output is driven by four variables:

1. $\sigma_{i,t}$ is the sectoral GDP share (composition state),
2. Energy-saving innovation ($\gamma \uparrow$), which is endogenous,
3. $p_{e,t}$ the level of the energy price,
4. r_t the interest rate (which governs capital and energy demand).

Endogenous Efficiency: Induced Innovation in the Model

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Exogenous energy pricing policy optimizes $E_t/Y_t \downarrow$ via two channels:

1. within-sector efficiency gains ($g_i \uparrow$)
2. reallocation toward more efficient sectors ($\sigma_{i,t}$ shifts).

see more

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Data

- **Scope:** Panel dataset for 25 Latin American countries, 1990–2021.
- **Variables:**
  - Sectoral GDP (CEPALSTAT): aggregated into 6 sectors (Agro-Fishing-Mining, Manufacturing, Construction, Transport, Hotels-Restaurants, Commerce-Services).
  - Sectoral energy consumption (OLADE).
  - Climate-policy records (Climate Policy Radar).
  - Control variables: population, trade openness, resource rents, emissions.
- **Derived variables:**
  - Sectoral GDP shares ($\sigma_{s,c,t} = Y_{s,c,t}/Y_{c,t}$) $\rightarrow$ structural-change channel.
  - Sectoral energy intensity ($e_{s,c,t} = E_{s,c,t}/Y_{s,c,t}$) $\rightarrow$ efficiency channel.
  - Aggregate energy intensity ($E_{c,t}/Y_{c,t} = \sum_s \sigma_{s,c,t} \times e_{s,c,t}$).

Descriptive Statistics

Variable		Mean	Median	Std. Dev.
		Latin American Countries		
Carbon intensity	(kt of CO2e/TJ)	0,176	0,133	0,179
Energy intensity	(TJ/$2018 PPP GDP)	11,250	8,324	8,520
Emissions per unit of GDP	(kt of CO2e/$2018 PPP GDP)	1,720	1,193	1,561
GDP	(US Millions, 2018)	119996,05	15057,75	303782,13
Number of climate policies approved		15,320	13,000	14,142
Agriculture, fishing and Mining	(share of GDP)	0,092	0,073	0,091
Manufacturing and Electricity, gas and water supply	(share of GDP)	0,195	0,191	0,072
Construction	(share of GDP)	0,072	0,066	0,029
Transport	(share of GDP)	0,100	0,093	0,032
Hotels and restaurants	(share of GDP)	0,031	0,028	0,033
Commercial, services, public	(share of GDP)	0,507	0,519	0,088
Agriculture, fishing and Mining	(TJ/$2018 PPP GDP)	15,101	3,931	30,440
Manufacturing and Electricity, gas and water supply	(TJ/$2018 PPP GDP)	15,767	9,651	15,957
Construction	(TJ/$2018 PPP GDP)	2,846	0,582	6,361
Transport	(TJ/$2018 PPP GDP)	34,681	32,971	20,328
Hotels and restaurants	(TJ/$2018 PPP GDP)	83,853	38,170	118,818
Commercial, services, public	(TJ/$2018 PPP GDP)	1,012	0,837	0,916

Empirical Strategies

Decomposition Analysis

- Energy intensity can be expressed in terms of energy consumption by sector, and sectoral change:

$$\frac{E_t}{Y_t} = \sum_i^N \frac{Y_{i,t}}{Y_t} \frac{e_{i,t}}{Y_{i,t}} = \sum_i^N \sigma_{i,t} \frac{e_{i,t}}{Y_{i,t}} \tag{2}$$

Empirical Exercises: Synthetic Control Method

- The effect of carbon tax on structural change and energy efficiency.

$$\hat{\tau}_{1t} = Y_{1t} - \sum_{j=2}^{J+1} \omega_j Y_{jt} \tag{3}$$

- Y_{1t}: Observed outcome in the treated unit (Mexico or Colombia) at time t.
- $Y_{1t}^N = \sum_{j=2}^{J+1} \omega_j Y_{jt}$: Counterfactual outcome in the absence of a carbon tax.
- Y_{jt}: Outcome of control unit j at time t.
- ω_j: Weight assigned to control unit j, selected to minimize pre-intervention distance.
- The causal effect $\hat{\tau}_{1t}$ is estimated as the deviation of the observed outcome from the counterfactual.

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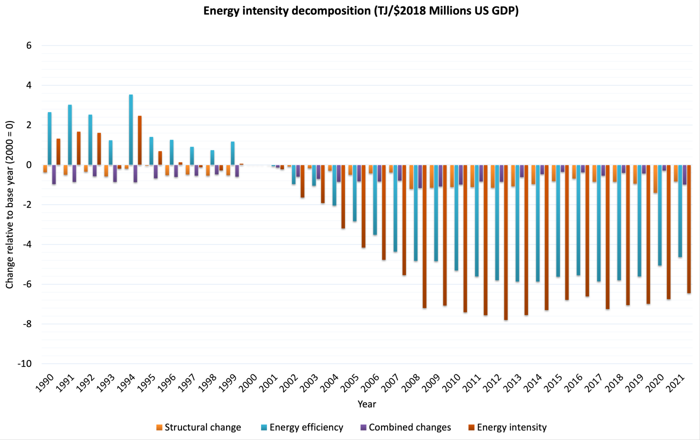
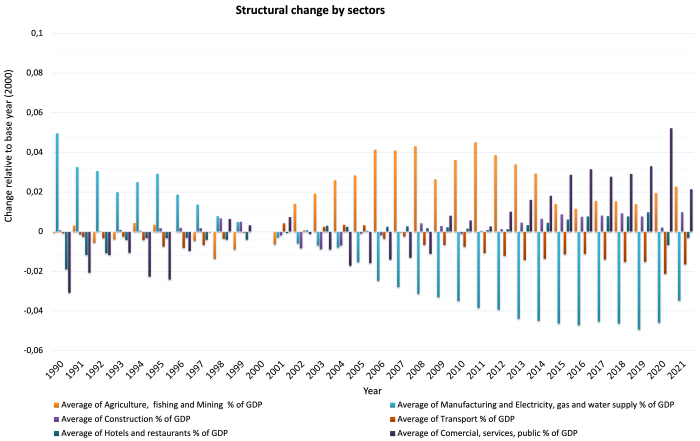
Energy Intensity Decomposition in Latin America



- Improvements in energy efficiency account for approximately 75% of the reduction in energy intensity.
- Structural change has contributed around 15% to the decline.

Structural Change in Latin America



- Declining GDP share (2000+): Manufacturing and transport.
- ↑ GDP share (2000+): Agriculture, mining, services, and construction.

Heterogeneities by income levels: View View

Energy Intensity Decomposition by Income

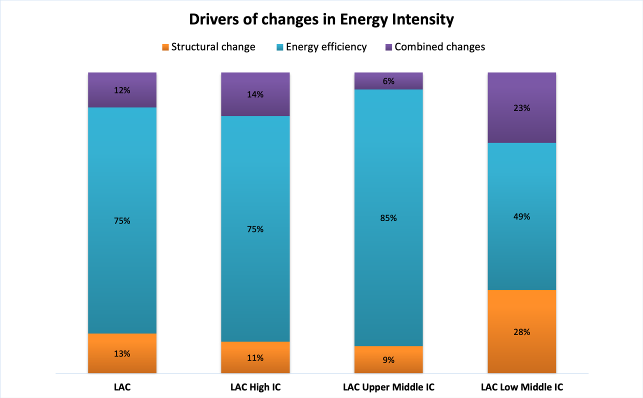


Figure: Components of energy intensity and changes by income from 2000-2021

- Impacts vary by income level - efficiency dominant for higher income countries, structure mattered more in the early 2000s for lower middle income.

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Structural Change by Income

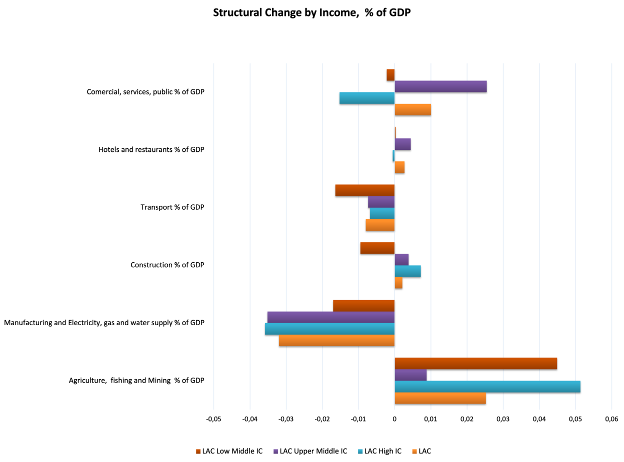


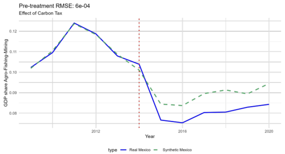
Figure: Changes in shares of GDP by sector from (2001 - 2021)

- In high-income countries, the structural shifts particularly benefit less

Empirical Results: Synthetic Controls

- Synthetic Control Results of **Sectoral GDP Share** for Mexico.
- Synthetic Control Results of **Sectoral Energy Intensity** for Mexico.
- Synthetic Control Results of **Sectoral GDP Share** for Colombia.
- Synthetic Control Results of **Sectoral Energy Intensity** for Colombia.
- Summary of the Results.

Synthetic Control Results of **Sectoral GDP Share** for Mexico

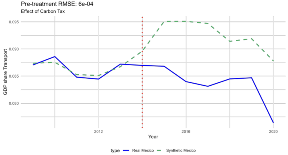


b. GDP share Agro-Fishing-Mining (%)

Note: The synthetic line represents Synthetic Mexico, built from a weighted combination of similar countries: Belize (0.2%), Brazil (38.6%), Ecuador (5.0%), Grenada (10.3%), Honduras (8.7%), Paraguay (0.4%), Peru (19.4%), and Suriname (17.5%).

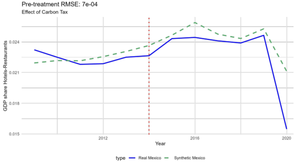
c. GDP share Manufacturing-Energy-Water (%)

Note: The synthetic line represents Synthetic Mexico, built from a weighted combination of similar countries: Cuba (70.0%) and Peru (30.0%).



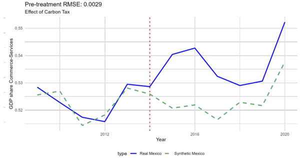
d. GDP share Transport (%)

Note: The synthetic line represents Synthetic Mexico, built from a weighted combination of similar countries: Brazil (0.1%), Costa Rica (35.0%), Cuba (1.2%), Haiti (10.2%), and Peru (53.5%).



e. GDP share Hotels and Restaurants (%)

Note: The synthetic line represents Synthetic Mexico, built from a weighted combination of similar countries: Bolivia (2.5%), Brazil (7.9%), Costa Rica (6.7%), El Salvador (7.1%), Haiti (16.0%), Honduras (6.1%), Panama (7.1%), Paraguay (6.6%), Suriname (6.2%), and Trinidad and Tobago



f. GDP share Commerce, Services (%)

Note: The synthetic line represents Synthetic Mexico, built from a weighted combination of similar countries: Belize (14.9%), Dominican Republic (21.9%), Haiti (3.0%), Honduras (54.8%), and Uruguay (5.4%).

Synthetic Control Results of **Sectoral Energy Intensity** for Mexico

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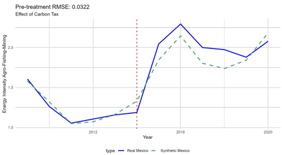
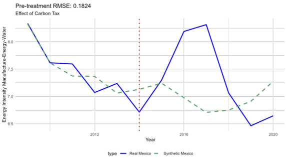


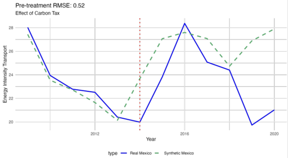
a. Agro-Fishing-Mining (TJ/$2018 PPP)

Note: The synthetic line represents Synthetic Mexico, built from a weighted combination of similar countries: Barbados (4.3%), Ecuador (45.2%), and Peru (50.4%).



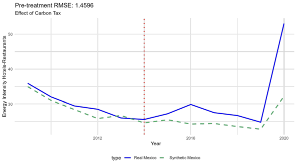
b. Manufacturing-Energy-Water (TJ/$2018 PPP)

Note: The synthetic line represents Synthetic Mexico, built from a weighted combination of similar countries: Ecuador (60.6%), Panama (4.8%), and Uruguay (34.6%).



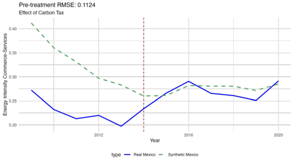
c. Transport (TJ/$2018 PPP)

Note: The synthetic line represents Synthetic Mexico, built from a weighted combination of similar countries: Brazil (5.0%), Haiti (40.9%), Peru (1.1%), and Trinidad and Tobago (53.0%).



d. Hotels and Restaurants (TJ/$2018 PPP)

Note: The synthetic line represents Synthetic Mexico, built from a weighted combination of similar countries: Grenada (0.5%), Peru (27.5%), and Trinidad and Tobago (72.0%).



e. Commerce, Services (TJ/$2018 PPP)

Note: The synthetic line represents Synthetic Mexico, built from a weighted combination of similar countries: Haiti (0.2%) and Trinidad and Tobago (99.8%).

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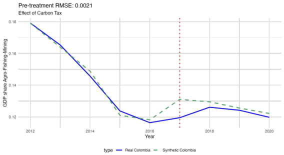
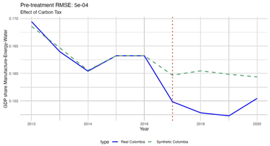
Synthetic Control Results of **Sectoral GDP Share** for Colombia



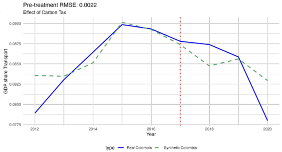
b. GDP share Agro-Fishing-Mining (%)

Note: The synthetic line represents Synthetic Colombia, built from a weighted combination of similar countries: Belize (14.7%), Ecuador (0.3%), Peru (43.9%), Suriname (22.3%), and Trinidad and Tobago (18.8%).



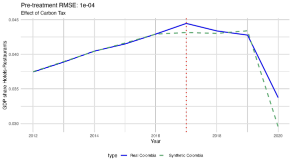
c. GDP share Manufacturing-Energy-Water (%)

Note: The synthetic line represents Synthetic Colombia, built from a weighted combination of similar countries: Bolivia (16.2%), Brazil (19.2%), Costa Rica (2.8%), Cuba (0.9%), Dominican Republic (10.1%), Honduras (8.4%), Nicaragua (0.9%), Peru (40.1%), Trinidad and Tobago (0.7%), and Uruguay (0.8%).



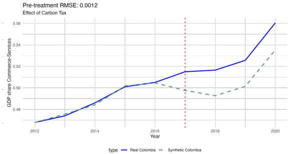
d. GDP share Transport (%)

Note: The synthetic line represents Synthetic Colombia, built from a weighted combination of similar countries: Brazil (8.5%), Costa Rica (3.6%), Cuba (1.3%), Paraguay (33.9%), and Peru (52.7%).



e. GDP share Hotels and Restaurants (%)

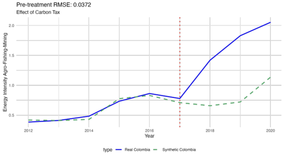
Note: The synthetic line represents Synthetic Colombia, built from a weighted combination of similar countries: Cuba (15.1%), El Salvador (35.1%), Panama (9.3%), and Peru (40.4%).



f. GDP share Commerce, Services (%)

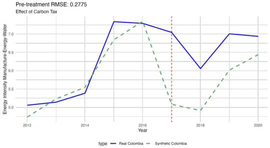
Note: The synthetic line represents Synthetic Colombia, built from a weighted combination of similar countries: Dominican Republic (0.1%), Ecuador (2.6%), Honduras (13.6%), Nicaragua (8.4%), Panama (17.7%), Paraguay (0.2%), Peru (27.1%), and Trinidad and Tobago (30.3%).

Synthetic Control Results of **Sectoral Energy Intensity** for Colombia



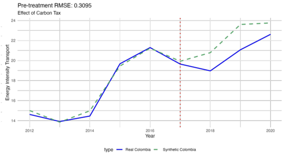
b. Energy Consumption Agro-Fishing-Mining

Note: The synthetic line represents Synthetic Colombia, built from a weighted combination of similar countries: Belize (39.5%), Bolivia (9.1%), and Ecuador (51.4%).



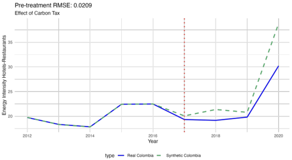
c. Energy Intensity Manufacturing-Energy-Water

Note: The synthetic line represents Synthetic Colombia, built from a weighted combination of similar countries: Ecuador (7.8%), Jamaica (44.3%), and Suriname (47.9%).



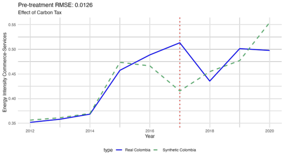
d. Energy Intensity Transport

Note: The synthetic line represents Synthetic Colombia, built from a weighted combination of similar countries: Dominican Republic (35.3%), Haiti (57.2%), and Peru (7.4%).



e. Energy Intensity Hotels and Restaurants

Note: The synthetic line represents Synthetic Colombia, built from a weighted combination of similar countries: Brazil (55.5%), Jamaica (13.6%), Peru (30.1%), and Suriname (0.8%).



f. Energy Intensity Commerce, and Services

Note: The synthetic line represents Synthetic Colombia, built from a weighted combination of similar countries: Brazil (85.0%) and Trinidad and Tobago (15.0%).

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Structural Change: GDP Shares after Carbon Tax

Mexico vs Colombia

Mexico (Significant Changes)

- Agro-Fishing-Mining: Decrease by 1.6 p.p. (significant)
- Construction: Decrease by 0.9 p.p. (significant)
- Transport: Decrease by 0.5 p.p. (significant)
- Manufacture-Energy-Water: Increase by 4.1 p.p. (highly significant)
- Commerce-Services: Increase by 1.5 p.p. (significant)

Colombia (Minor Changes)

- No statistically significant changes, except Construction (almost significant, small decrease).
- Sectoral composition largely unaffected.

Conclusion: Mexico shows clear structural shift toward less carbon-intensive sectors; Colombia shows economic resilience or limited sectoral response.

Energy Efficiency Changes after Carbon Tax

Mexico vs Colombia

Mexico (Mixed Results)

- Transport: -11.4% (strong improvement)
- Commerce-Services: -5% (small improvement)
- Agro-Fishing-Mining: +9% (worsening)
- Manufacturing and Hotels: Moderate worsening (+5%-6%)

Colombia (Severe in Agriculture)

- Agro-Fishing-Mining: +109% (severe worsening)
- Hotels-Restaurants: -15.6% (significant improvement)
- Transport: -10.2% (moderate improvement)
- Manufacturing and Commerce: Increase in energy intensity.

Conclusion: Mexico achieved sector-specific improvements; Colombia improved mainly in transport and hotels but suffered severe losses in agriculture.

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Conclusions

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- Climate mitigation policies can provide incentives that facilitate structural shifts toward less energy-intensive economic activities and promote greater energy efficiency improvements across sectors
- However, Relying solely on price-based instruments (e.g., carbon taxes) is suboptimal in developing, resource-dependent systems due to the perverse scale effects in heavy sectors.
- There is significant variation in sectoral impacts:
  - Emissions-intensive sectors like agriculture, mining and manufacturing, and transport may see declines in economic activity.
  - Meanwhile, services sector may benefit.
  - Transport sector reduces energy intensity.
- Sector-level effects on structure and efficiency changes are heterogeneous

Thank You & Discussion

Thank you for your attention.

Questions & Discussion

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Decision-Maker Problem

Representative decision-maker:

- Maximizes a utility (objective) function subject to a budget constraint:

$$\max_{c_{i,t}} \sum_{i=1}^{N} \lambda_i \; u(c_{i,t}) \quad \text{s.t.} \quad a_t r_t + w_t + p_{e,t}\bar{e}_t = \sum_{i=1}^{N} p_{i,t} c_{i,t}$$

First-order condition — ratio of expenditure between any pair of goods:

$$\frac{u'(c_j)}{u'(c_i)} = \frac{p_{j,t}}{p_{i,t}} \frac{\lambda_i}{\lambda_j} \tag{4}$$

Assumption 1: Elasticity of substitution between goods > 1.

Lemma

For any pair of goods i, j, an increase in the relative price of good i generates a reallocation of expenditure in favor of good j and against good i:

$$\frac{\partial\left(\frac{p_{i,t}c_{i,t}}{p_{j,t}c_{j,t}}\right)}{\partial\left(\frac{p_{i,t}}{p_{j,t}}\right)} < 0$$

Producer (Optimizing-Unit) Problem

Capital and energy are *perfect complements* (Leontief constraint); this input combines with labor through a Cobb-Douglas production function.

Producer optimization program:

$$\max_{K_{i,t},L_{i,t}} \left\{p_{i,t}A_{i,t}\left(\min\left[K_{i,t},\gamma_{i,t}e_{i,t}\right]\right)^{\alpha} L_{i,t}^{1-\alpha} - r_t K_{i,t} - w_t L_{i,t} - p_{e,t}e_{i,t}\right\}$$

$$\max_{K_{i,t},L_{i,t}} \left\{p_{i,t}A_{i,t}(K_{i,t})^{\alpha}(L_{i,t})^{1-\alpha} - r_t K_{i,t} - w_t L_{i,t} - p_{e,t}\frac{K_{i,t}}{\gamma_{i,t}}\right\}$$

Optimal input relations:

$$\frac{k_{j,t}}{k_{i,t}} = \left(\frac{r_t + \frac{p_{e,t}}{\gamma_{i,t}}}{r_t + \frac{p_{e,t}}{\gamma_{j,t}}}\right) \tag{5}$$

$$\frac{p_{i,t}}{p_{j,t}} = \frac{A_{j,t}}{A_{i,t}}\left(\frac{r_t + \frac{p_{e,t}}{\gamma_{i,t}}}{r_t + \frac{p_{e,t}}{\gamma_{j,t}}}\right)^{\alpha} \tag{6}$$

- The optimal capital-labor ratio of a sector falls if the effective cost of operating capital with energy rises.
- The relative price between two goods is a function of relative productivity and relative energy cost.

Solving the System: General Equilibrium

Consumption relation (optimality condition):

$$\frac{u'(c_{i,t})}{u'(c_{j,t})} = \frac{\lambda_j}{\lambda_i} \frac{A_{j,t}}{A_{i,t}} \left(\frac{r_t + \frac{p_{e,t}}{\gamma_{i,t}}}{r_t + \frac{p_{e,t}}{\gamma_{j,t}}} \right)^{\alpha} \tag{7}$$

Lemma

For any pair of goods i, j, where j is more energy-efficient than i ($\gamma_{j,t} > \gamma_{i,t}$), an increase in the energy price $p_{e,t}$ reallocates consumption toward j and away from i:

$$\frac{\partial \left(\frac{c_{i,t}}{c_{j,t}} \right)}{\partial p_{e,t}} < 0$$

Corollary 1:

$$\frac{\partial \left(\frac{p_{i,t} Y_{i,t}}{p_{j,t} Y_{j,t}} \right)}{\partial \left(\frac{p_{i,t}}{p_{j,t}} \right)} < 0$$

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Decoupling Energy from Output: The Objective Function

Corollary

Holding the rest of the system constant, an increase in the energy price reduces the equilibrium interest rate: $\frac{\partial(r_t)}{\partial p_{e,t}} = -\frac{1}{\gamma_t}$.

System-level energy intensity (the metric being optimized):

$$\frac{E_t}{Y_t} = \alpha \sum_{i=1}^{N} \frac{\sigma_{i,t}}{\gamma_{i,t} r_t + p_{e,t}}$$

Lemma

Aggregate energy intensity is a function of:

- *energy-saving innovation (state variable* γ*),*
- *the energy price (control variable),*
- *sectoral shares* $\sigma_{i,t} = \frac{p_{i,t} Y_{i,t}}{Y_t}$ *(allocation variables),*
- *the interest rate (equilibrium feedback variable).*

Sensitivity Analysis: Comparative Statics of the System

Partial-derivative (sensitivity) results:

- $\frac{\partial(E_t/Y_t)}{\partial\gamma_{i,t}} = -\alpha r_t \frac{\sigma_{i,t}}{(\gamma_{i,t}r_t+p_{e,t})^2} < 0$
- $\frac{\partial(E_t/Y_t)}{\partial p_{e,t}} = -\alpha \sum \frac{\sigma_{i,t}}{(\gamma_{i,t}r_t+p_{e,t})^2} < 0$
- $\frac{\partial(E_t/Y_t)}{\partial\sigma_{i,t}} = \alpha \frac{1}{\gamma_{i,t}r_t+p_{e,t}} > 0$
- $\frac{\partial(E_t/Y_t)}{\partial r_t} = -\sum \frac{\sigma_{i,t}\gamma_{i,t}}{(\gamma_{i,t}r_t+p_{e,t})^2} < 0$

Interpretation: the system's energy intensity responds to (1) innovation, (2) price, (3) reallocation, and (4) interest-rate feedback — the four levers a policy-engineering exercise must jointly account for.

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Simulated Effect of a Carbon-Tax Control Variable

Total-derivative (system-level) effect:

$$\frac{\partial(E_t/Y_t)}{\partial\tau} = \alpha \left\{ \begin{aligned} &\sum_{i=1}^{N} \frac{\sigma_{i,t}}{(\gamma_{i,t}r_t + p_{e,t})^2} \left(\frac{\gamma_{i,t}}{\gamma_t} - 1 \right) \frac{\partial p_{e,t}}{\partial\tau} \\ &- r_t \sum_{i=1}^{N} \frac{\sigma_{i,t}}{(\gamma_{i,t}r_t + p_{e,t})^2} \frac{\partial\gamma_{i,t}}{\partial\tau} \\ &+ \sum_{i=1}^{N} \frac{1}{\gamma_{i,t}r_t + p_{e,t}} \frac{\partial\sigma_{i,t}}{\partial\tau} \end{aligned} \right\} \tag{8}$$

Mechanisms driving the optimum:

- $\tau \Rightarrow \uparrow p_e \Rightarrow \downarrow e \Rightarrow \downarrow k \Rightarrow \downarrow r \Rightarrow \uparrow k$. The net effect on each sector depends on its energy efficiency relative to the system-wide average.
- $\uparrow p_e \Rightarrow \uparrow \gamma \Rightarrow \uparrow Y$ while E stays constant — an induced energy-saving-innovation channel.
- Sectoral output is reallocated toward more energy-efficient nodes.

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